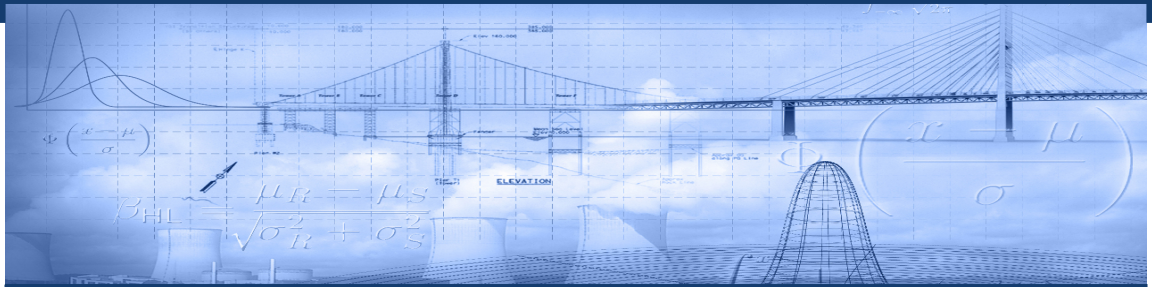


PCE-based Sobol' analysis

N. Lüthen, B. Sudret

Chair of Risk, Safety and Uncertainty Quantification, ETH Zürich, Switzerland

Outline

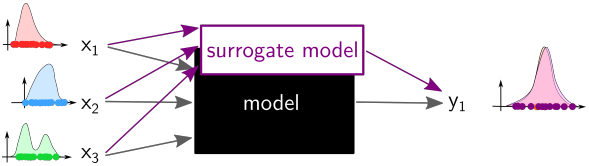
Surrogate modelling

Polynomial chaos expansion

Sobol' indices defined based on PCE

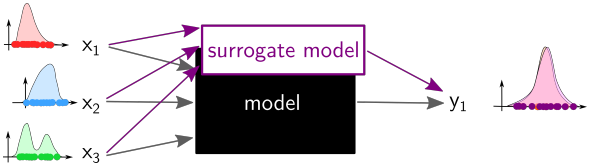
Surrogate modelling

Replace the expensive original model by an accurate but cheaper approximation



Surrogate modelling

Replace the expensive original model by an accurate but cheaper approximation

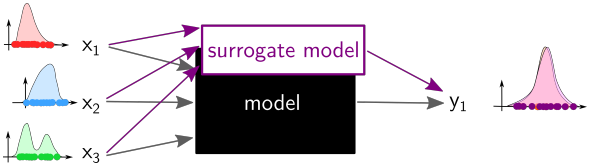


Popular surrogate models in engineering:

- Gaussian processes / Kriging
- Polynomial response surfaces / polynomial chaos expansions
- Support vector machines (SVM), random forests, neural networks...

Surrogate modelling

Replace the expensive original model by an accurate but cheaper approximation



Popular surrogate models in engineering:

- Gaussian processes / Kriging
- Polynomial response surfaces / polynomial chaos expansions
- Support vector machines (SVM), random forests, neural networks...

Used for

- Uncertainty propagation
- Sensitivity analysis
- Optimization
- Reliability analysis
- ...

Setting

Model

Output

Input

$$Y = \mathcal{M}(\boldsymbol{X})$$

- $\boldsymbol{X}$ is a d-dimensional random vector with
  - independent components
  - density $f_{\boldsymbol{X}} = \prod_{i=1}^{d} f_{X_i}$
  - finite variance
  - values in $\mathcal{D} \subseteq \mathbb{R}^d$

Setting

Output Model Input

$$Y = \mathcal{M}(\boldsymbol{X})$$

- $\boldsymbol{X}$ is a d-dimensional random vector with
  - independent components
  - density $f_{\boldsymbol{X}} = \prod_{i=1}^{d} f_{X_i}$
  - finite variance
  - values in $\mathcal{D} \subseteq \mathbb{R}^d$
- $\mathcal{M}$ is a function in $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- $Y = \mathcal{M}(\boldsymbol{X})$ is a random variable with finite variance

Setting

Output Model Input

$$Y = \mathcal{M}(\boldsymbol{X})$$

- $\boldsymbol{X}$ is a d-dimensional random vector with
  - independent components
  - density $f_{\boldsymbol{X}} = \prod_{i=1}^{d} f_{X_i}$
  - finite variance
  - values in $\mathcal{D} \subseteq \mathbb{R}^d$
- $\mathcal{M}$ is a function in $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- $Y = \mathcal{M}(\boldsymbol{X})$ is a random variable with finite variance

We want to study the random variable Y

Setting

Model

Output Input

$$Y = \mathcal{M}(\boldsymbol{X})$$

- $\boldsymbol{X}$ is a d-dimensional random vector with
  - independent components
  - density $f_{\boldsymbol{X}} = \prod_{i=1}^{d} f_{X_i}$
  - finite variance
  - values in $\mathcal{D} \subseteq \mathbb{R}^d$
- $\mathcal{M}$ is a function in $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- $Y = \mathcal{M}(\boldsymbol{X})$ is a random variable with finite variance

We want to study the random variable Y

We want to approximate the mapping $\mathcal{M}$

Illustration: Univariate polynomial bases

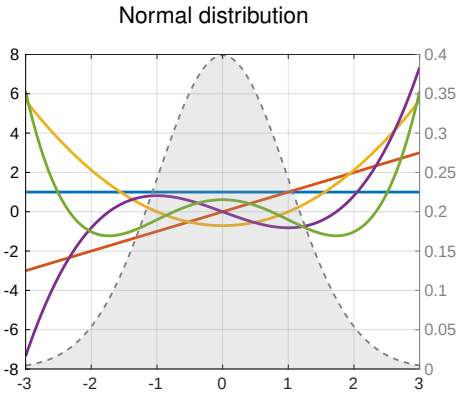
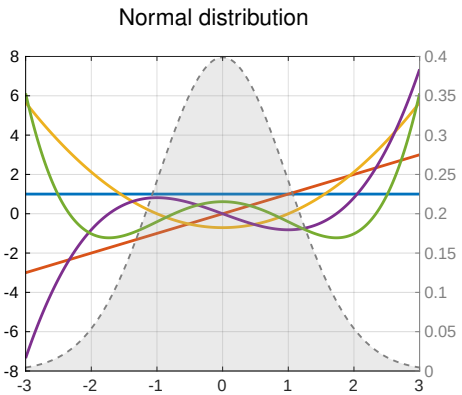
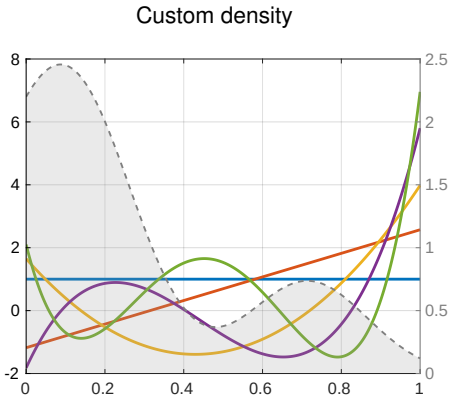


Illustration: Univariate polynomial bases





Polynomial chaos expansion

Model

Output

Input

$$Y = \mathcal{M}(\boldsymbol{X}) \quad = \quad \mathcal{M}^{\mathsf{PCE}}(\boldsymbol{X}) = \sum_{\boldsymbol{\alpha} \in \mathbb{N}^d} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

Polynomial chaos expansion

Output Model Input Surrogate model Coefficients Basis function

$$Y = \mathcal{M}(\boldsymbol{X}) \quad = \quad \mathcal{M}^{\text{PCE}}(\boldsymbol{X}) = \sum_{\boldsymbol{\alpha} \in \mathbb{N}^d} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

Polynomial chaos expansion

Model

Surrogate model

Output

Input

Coefficients

$$Y = \mathcal{M}(\boldsymbol{X}) \quad = \quad \mathcal{M}^{\text{PCE}}(\boldsymbol{X}) = \sum_{\boldsymbol{\alpha} \in \mathbb{N}^d} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

Basis function

where $\{\psi_{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha} \in \mathbb{N}^d}$ ...

- is a basis of $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- consists of multivariate polynomials $\psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) = \prod_{i=1}^{d} \psi^{(i)}_{\alpha_i}(x_i)$
- is orthonormal wrt $f_{\boldsymbol{X}}$,

Polynomial chaos expansion

Model

Surrogate model

Output

Input

Coefficients

$$Y = \mathcal{M}(\boldsymbol{X}) \approx \mathcal{M}(\boldsymbol{X}) = \sum_{\boldsymbol{\alpha} \in \mathcal{A}} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

Basis function

where $\{\psi_{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha} \in \mathbb{N}^d}$ ...

- is a basis of $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- consists of multivariate polynomials $\psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) = \prod_{i=1}^{d} \psi^{(i)}_{\alpha_i}(x_i)$
- is orthonormal wrt $f_{\boldsymbol{X}}$,

$\alpha \in \mathbb{N}^d$ encodes the degree of $\psi_{\boldsymbol{\alpha}}$ in each dimension,

Polynomial chaos expansion

Model

Surrogate model

Output

Input

Coefficients

$$Y = \mathcal{M}(\boldsymbol{X}) \approx \mathcal{M}(\boldsymbol{X}) = \sum_{\boldsymbol{\alpha} \in \mathcal{A}} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

Basis function

where $\{\psi_{\boldsymbol{\alpha}}\}_{\boldsymbol{\alpha} \in \mathbb{N}^d}$ ...

- is a basis of $L^2_{f_{\boldsymbol{X}}}(\mathcal{D})$
- consists of multivariate polynomials $\psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) = \prod_{i=1}^{d} \psi^{(i)}_{\alpha_i}(x_i)$
- is orthonormal wrt $f_{\boldsymbol{X}}$,

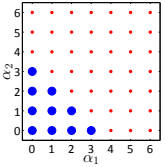
$\alpha \in \mathbb{N}^d$ encodes the degree of $\psi_{\boldsymbol{\alpha}}$ in each dimension,
and $\mathcal{A} \subset \mathbb{N}^d$ is the set of multi-indices, e.g., total-degree basis of degree p:

$$\mathcal{A} = \{\boldsymbol{\alpha} \in \mathbb{N}^d : \sum_{i=1}^{d} \alpha_i \leq p\}$$

Example in $d = 2$ dimensions

- Model $Y = \mathcal{M}(X_1, X_2)$
- Gaussian distribution $\rightarrow$ Hermite polynomials
- Total degree $\leq p = 3$

j	$\boldsymbol{\alpha}$	$\Psi_{\boldsymbol{\alpha}} \equiv \Psi_j$
0	$[0,0]$	$\Psi_0(\xi_1,\xi_2) = 1$
1	$[1,0]$	$\Psi_1(\xi_1,\xi_2) = \xi_1$
2	$[0,1]$	$\Psi_2(\xi_1,\xi_2) = \xi_2$
3	$[2,0]$	$\Psi_3(\xi_1,\xi_2) = (\xi_1^2 - 1)/\sqrt{2}$
4	$[1,1]$	$\Psi_4(\xi_1,\xi_2) = \xi_1\xi_2$
5	$[0,2]$	$\Psi_5(\xi_1,\xi_2) = (\xi_2^2 - 1)/\sqrt{2}$
6	$[3,0]$	$\Psi_6(\xi_1,\xi_2) = (\xi_1^3 - 3\xi_1)/\sqrt{6}$
7	$[2,1]$	$\Psi_7(\xi_1,\xi_2) = (\xi_1^2 - 1)\xi_2/\sqrt{2}$
8	$[1,2]$	$\Psi_8(\xi_1,\xi_2) = (\xi_2^2 - 1)\xi_1/\sqrt{2}$
9	$[0,3]$	$\Psi_9(\xi_1,\xi_2) = (\xi_2^3 - 3\xi_2)/\sqrt{6}$



$$\text{card}\mathcal{A}^{2,3} = \frac{(2+3)!}{2!\,3!} = 10$$

$$\begin{aligned}\tilde{Y} \equiv \mathcal{M}^{\text{PC}}(\xi_1,\xi_2) &= c_0 + c_1\,\xi_1 + c_2\,\xi_2 \\ &+ c_3\,(\xi_1^2 - 1)/\sqrt{2} + c_4\,\xi_1\xi_2 \\ &+ c_5\,(\xi_2^2 - 1)/\sqrt{2} + c_6\,(\xi_1^3 - 3\xi_1)/\sqrt{6} \\ &+ c_7\,(\xi_1^2 - 1)\xi_2/\sqrt{2} + c_8\,(\xi_2^2 - 1)\xi_1/\sqrt{2} \\ &+ c_9(\xi_2^3 - 3\xi_2)/\sqrt{6}\end{aligned}$$

Computation of polynomial chaos expansion

$$Y \approx \sum_{\boldsymbol{\alpha} \in \mathcal{A}} c_{\boldsymbol{\alpha}} \psi_{\boldsymbol{\alpha}}(\boldsymbol{X})$$

- Assemble the linear system:

$$\boldsymbol{\Psi} \boldsymbol{c} \approx \boldsymbol{y}$$

  with the regression matrix $\Psi_{ij} = \psi_{\boldsymbol{\alpha}_j}(\boldsymbol{x}^{(i)})$ and the right-hand side vector $y_i = \mathcal{M}(\boldsymbol{x}^{(i)})$
- Solve e.g. by ℓ^1-minimization:

$$\min_{\boldsymbol{c}} \left(\|\boldsymbol{\Psi} \boldsymbol{c} - \boldsymbol{y}\|_2^2 + \gamma \|\boldsymbol{c}\|_1 \right)$$

- ... or other sparse regression techniques based on a small number of model evaluations

Lüthen, Marelli, Sudret (2021)

Reminder: Orthonormality of PCE basis functions

PCE basis functions are orthonormal w.r.t. $f_{\boldsymbol{X}}$. This means:

- They are orthogonal: For $\psi_{\boldsymbol{\alpha}}, \psi_{\boldsymbol{\beta}}$ with $\boldsymbol{\alpha} \neq \boldsymbol{\beta}$, it holds that

$$0 = \langle \psi_{\boldsymbol{\alpha}}, \psi_{\boldsymbol{\beta}} \rangle_{f_{\boldsymbol{X}}} = \int \psi_{\boldsymbol{\alpha}}(\boldsymbol{x}) \psi_{\boldsymbol{\beta}}(\boldsymbol{x})\, f_{\boldsymbol{X}}(\boldsymbol{x})\, \mathrm{d}x = \mathbb{E}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X}) \psi_{\boldsymbol{\beta}}(\boldsymbol{X})\right]$$

- They have unit norm:

$$1 = \langle \psi_{\boldsymbol{\alpha}}, \psi_{\boldsymbol{\alpha}} \rangle_{f_{\boldsymbol{X}}} = \mathbb{E}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})^2\right]$$

It follows that:

- Since $\psi_{\boldsymbol{0}} = 1$ is constant, all other basis functions are centered, i.e. for $\boldsymbol{\alpha} \neq \boldsymbol{0}$ we have

$$0 = \langle \psi_{\boldsymbol{\alpha}}, \psi_{\boldsymbol{0}} \rangle_{f_{\boldsymbol{X}}} = \langle \psi_{\boldsymbol{\alpha}}, 1 \rangle_{f_{\boldsymbol{X}}} = \int \psi_{\boldsymbol{\alpha}}(\boldsymbol{x})\, f_{\boldsymbol{X}}(\boldsymbol{x})\, \mathrm{d}x = \mathbb{E}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right]$$

- Unit variance: $\mathrm{Var}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = 1$
- Two random variables $\psi_{\boldsymbol{\alpha}}(\boldsymbol{X}), \psi_{\boldsymbol{\beta}}(\boldsymbol{X})$ with $\boldsymbol{\alpha} \neq \boldsymbol{\beta}$ are uncorrelated

Postprocessing of PCE

- The response moments can be computed analytically:

$$\mathbb{E}\left[Y\right] = \mathbb{E}\left[\sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\mathbb{E}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = c_{\boldsymbol{0}}$$

$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\mathrm{Var}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\neq\boldsymbol{0}} c_{\boldsymbol{\alpha}}^2$$

Postprocessing of PCE

- The response moments can be computed analytically:

$$\mathbb{E}\left[Y\right] = \mathbb{E}\left[\sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\mathbb{E}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = c_{\boldsymbol{0}}$$

$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\in\mathbb{N}} c_{\boldsymbol{\alpha}}\mathrm{Var}\left[\psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}\neq\boldsymbol{0}} c_{\boldsymbol{\alpha}}^2$$

- Also: we get Sobol' indices and the ANOVA decomposition analytically!

Outline

Surrogate modelling

Polynomial chaos expansion

Sobol' indices defined based on PCE
- Variance decomposition for PCE
- Sobol' indices

Activity: What does a PCE tell us about sensitivity?

Assume that you have computed a PCE for your model and you obtained the sparse representation (multi-indices and corresponding nonzero coefficients) gathered in the table below.

j	Multi-index α	Coefficient value
0	[0 0 0]	7.2
1	[0 1 0]	3.4
2	[0 0 1]	-0.9
3	[0 1 1]	0.1
4	[0 2 0]	1.7
5	[2 1 0]	-0.02
6	[0 3 0]	-0.2

What do the coefficients tell us about which variables are important or unimportant?

Hoeffding-Sobol decomposition via PCE

Let $\mathcal{M}^{\text{PCE}}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$ be a PCE model.

We group the PCE terms according to the variables involved:

$$g_0 = c_{(0,\dots,0)}$$

Hoeffding-Sobol decomposition via PCE

Let $\mathcal{M}^{\text{PCE}}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$ be a PCE model.

We group the PCE terms according to the variables involved:

$$g_0 = c_{(0,\dots,0)}$$

$$g_1(x_1) = c_{(1,0,\dots,0)}\Psi_{(1,0,\dots,0)}(x_1) + c_{(2,0,\dots,0)}\Psi_{(2,0,\dots,0)}(x_1) + c_{(3,0,\dots,0)}\Psi_{(3,0,\dots,0)}(x_1) + \dots$$

Hoeffding-Sobol decomposition via PCE

Let $\mathcal{M}^{\text{PCE}}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$ be a PCE model.

We group the PCE terms according to the variables involved:

$$g_0 = c_{(0,\dots,0)}$$

$$g_1(x_1) = c_{(1,0,\dots,0)}\Psi_{(1,0,\dots,0)}(x_1) + c_{(2,0,\dots,0)}\Psi_{(2,0,\dots,0)}(x_1) + c_{(3,0,\dots,0)}\Psi_{(3,0,\dots,0)}(x_1) + \dots$$

$$g_i(x_i) = \sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

Hoeffding-Sobol decomposition via PCE

Let $\mathcal{M}^{\text{PCE}}(\boldsymbol{x}) = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$ be a PCE model.

We group the PCE terms according to the variables involved:

$$g_0 = c_{(0,\dots,0)}$$

$$g_1(x_1) = c_{(1,0,\dots,0)}\Psi_{(1,0,\dots,0)}(x_1) + c_{(2,0,\dots,0)}\Psi_{(2,0,\dots,0)}(x_1) + c_{(3,0,\dots,0)}\Psi_{(3,0,\dots,0)}(x_1) + \dots$$

$$g_i(x_i) = \sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}}\Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

$$g_{ij}(x_i, x_j) = \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_j>0 \\ \text{and } \alpha_k=0, k\neq i,j}} c_{\boldsymbol{\alpha}}\Psi_{\boldsymbol{\alpha}}(\boldsymbol{x})$$

$$\vdots$$

Example in $d = 2$ dimensions

- Model $Y = \mathcal{M}(X_1, X_2)$
- Gaussian distribution $\rightarrow$ Hermite polynomials
- Total degree $\leq p = 3$

j	$\boldsymbol{\alpha}$	$\Psi_{\boldsymbol{\alpha}} \equiv \Psi_j$
0	$[0,0]$	$\Psi_0(\xi_1,\xi_2) = 1$
1	$[1,0]$	$\Psi_1(\xi_1,\xi_2) = \xi_1$
2	$[0,1]$	$\Psi_2(\xi_1,\xi_2) = \xi_2$
3	$[2,0]$	$\Psi_3(\xi_1,\xi_2) = (\xi_1^2 - 1)/\sqrt{2}$
4	$[1,1]$	$\Psi_4(\xi_1,\xi_2) = \xi_1\xi_2$
5	$[0,2]$	$\Psi_5(\xi_1,\xi_2) = (\xi_2^2 - 1)/\sqrt{2}$
6	$[3,0]$	$\Psi_6(\xi_1,\xi_2) = (\xi_1^3 - 3\xi_1)/\sqrt{6}$
7	$[2,1]$	$\Psi_7(\xi_1,\xi_2) = (\xi_1^2 - 1)\xi_2/\sqrt{2}$
8	$[1,2]$	$\Psi_8(\xi_1,\xi_2) = (\xi_2^2 - 1)\xi_1/\sqrt{2}$
9	$[0,3]$	$\Psi_9(\xi_1,\xi_2) = (\xi_2^3 - 3\xi_2)/\sqrt{6}$

PCE

$$\begin{aligned}\tilde{Y} \equiv \mathcal{M}^{\mathrm{PC}}(\xi_1,\xi_2) &= c_{[0,0]} + c_{[1,0]}\,\xi_1 + c_{[0,1]}\,\xi_2 \\ &+ c_{[2,0]}\,(\xi_1^2 - 1)/\sqrt{2} + c_{[1,1]}\,\xi_1\xi_2 \\ &+ c_{[0,2]}\,(\xi_2^2 - 1)/\sqrt{2} + c_{[3,0]}\,(\xi_1^3 - 3\xi_1)/\sqrt{6} \\ &+ c_{[2,1]}\,(\xi_1^2 - 1)\xi_2/\sqrt{2} + c_{[1,2]}\,(\xi_2^2 - 1)\xi_1/\sqrt{2} \\ &+ c_{[0,3]}(\xi_2^3 - 3\xi_2)/\sqrt{6}\end{aligned}$$

$\rightarrow$ Sobol-Hoeffding decomposition

$$\begin{aligned}g_0 &= c_{[0,0]} \\ g_1(\xi_1) &= c_{[1,0]}\,\xi_1 + c_{[2,0]}\,(\xi_1^2 - 1)/\sqrt{2} \\ &\quad + c_{[3,0]}\,(\xi_1^3 - 3\xi_1)/\sqrt{6} \\ g_2(\xi_2) &= c_{[0,1]}\,\xi_2 + c_{[0,2]}\,(\xi_2^2 - 1)/\sqrt{2} \\ &\quad + c_{[0,3]}\,(\xi_2^2 - 1)\xi_1/\sqrt{2} \\ g_{12}(\xi_1,\xi_2) &= c_{[1,1]}\,\xi_1\xi_2 + c_{[2,1]}\,(\xi_1^2 - 1)\xi_2/\sqrt{2} \\ &\quad + c_{[1,2]}(\xi_2^3 - 3\xi_2)/\sqrt{6}\end{aligned}$$

Variance decomposition for PCE

From orthonormality of the PCE basis functions, it follows that...

Total variance

$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\mathcal{M}(\boldsymbol{X})\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}}^2 \mathrm{Var}\left[\Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}:\boldsymbol{\alpha}\neq\boldsymbol{0}} c_{\boldsymbol{\alpha}}^2$$

Variance decomposition for PCE

From orthonormality of the PCE basis functions, it follows that...

Total variance

$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\mathcal{M}(\boldsymbol{X})\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}}^2 \mathrm{Var}\left[\Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}:\boldsymbol{\alpha}\neq\boldsymbol{0}} c_{\boldsymbol{\alpha}}^2$$

Partial variances

$$\mathrm{Var}\left[\mathcal{M}_i(\boldsymbol{X})\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}}^2$$

Variance decomposition for PCE

From orthonormality of the PCE basis functions, it follows that...

Total variance

$$\mathrm{Var}\left[Y\right] = \mathrm{Var}\left[\mathcal{M}(\boldsymbol{X})\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}} c_{\boldsymbol{\alpha}}^2 \mathrm{Var}\left[\Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}:\boldsymbol{\alpha}\neq\boldsymbol{0}} c_{\boldsymbol{\alpha}}^2$$

Partial variances

$$\mathrm{Var}\left[\mathcal{M}_i(\boldsymbol{X})\right] = \mathrm{Var}\left[\sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}} \Psi_{\boldsymbol{\alpha}}(\boldsymbol{X})\right] = \sum_{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_k=0, k\neq i} c_{\boldsymbol{\alpha}}^2$$

$$\mathrm{Var}\left[\mathcal{M}_{ij}(\boldsymbol{X})\right] = \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_j>0 \\ \text{and } \alpha_k=0, k\neq i,j}} c_{\boldsymbol{\alpha}}^2$$

etc

Sobol' indices

Sobol' indices are the ratios of partial variances to the total variance $D = \mathrm{Var}\left[Y\right]$:

- First-order

$$S_i = \frac{\mathrm{Var}\left[\mathcal{M}_i(\boldsymbol{X})\right]}{\mathrm{Var}\left[Y\right]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \\ \text{and } \alpha_k=0, k\neq i}} c_{\boldsymbol{\alpha}}^2$$

Sobol' indices

Sobol' indices are the ratios of partial variances to the total variance $D = \mathrm{Var}\left[Y\right]$:

- First-order

$$S_i = \frac{\mathrm{Var}\left[\mathcal{M}_i(\boldsymbol{X})\right]}{\mathrm{Var}\left[Y\right]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \\ \text{and } \alpha_k=0, k\neq i}} c_{\boldsymbol{\alpha}}^2$$

- Second-order

$$S_{ij} = \frac{\mathrm{Var}\left[M_{ij}(\boldsymbol{X})\right]}{\mathrm{Var}\left[Y\right]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_j>0 \\ \text{and } \alpha_k=0, k\neq i,j}} c_{\boldsymbol{\alpha}}^2$$

Sobol' indices

Sobol' indices are the ratios of partial variances to the total variance $D = \mathrm{Var}\,[Y]$:

- First-order

$$S_i = \frac{\mathrm{Var}\,[\mathcal{M}_i(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \\ \text{and } \alpha_k=0, k\neq i}} c_{\boldsymbol{\alpha}}^2$$

- Second-order

$$S_{ij} = \frac{\mathrm{Var}\,[M_{ij}(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_j>0 \\ \text{and } \alpha_k=0, k\neq i,j}} c_{\boldsymbol{\alpha}}^2$$

- k-th order: $S_{i_1\ldots i_k} = \dfrac{\mathrm{Var}\,[\mathcal{M}_{i_1\ldots i_k}(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} =$

  Sum of all squared coefficients involving $i_1, \ldots, i_k$ (and nothing else) divided by D

Sobol' indices

Sobol' indices are the ratios of partial variances to the total variance $D = \mathrm{Var}\,[Y]$:

- First-order

$$S_i = \frac{\mathrm{Var}\,[\mathcal{M}_i(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \\ \text{and } \alpha_k=0, k\neq i}} c_{\boldsymbol{\alpha}}^2$$

- Second-order

$$S_{ij} = \frac{\mathrm{Var}\,[M_{ij}(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} = \frac{1}{D} \sum_{\substack{\boldsymbol{\alpha}:\alpha_i>0 \text{ and } \alpha_j>0 \\ \text{and } \alpha_k=0, k\neq i,j}} c_{\boldsymbol{\alpha}}^2$$

- k-th order: $S_{i_1\ldots i_k} = \dfrac{\mathrm{Var}\,[\mathcal{M}_{i_1\ldots i_k}(\boldsymbol{X})]}{\mathrm{Var}\,[Y]} =$
  Sum of all squared coefficients involving $i_1, \ldots, i_k$ (and nothing else) divided by D

Proportion of variance explained by X_i alone /
by X_i and X_j together / by $X_{i_1}, \ldots, X_{i_k}$ together

Properties of Sobol' indices

- There are $2^M - 1$ different indices including:
  - M first-order indices
  - $\binom{M}{2} = \frac{M(M-1)}{2}$ second-order indices
  - $\binom{M}{k}$ k-th order indices
- The sum of all Sobol' indices (of any order) is 1

Properties of Sobol' indices

- There are $2^M - 1$ different indices including:
  - M first-order indices
  - $\binom{M}{2} = \frac{M(M-1)}{2}$ second-order indices
  - $\binom{M}{k}$ k-th order indices
- The sum of all Sobol' indices (of any order) is 1

First-order indices

- $\sum_{i=1}^{M} S_i \leq 1$

Properties of Sobol' indices

- There are $2^M - 1$ different indices including:
  - M first-order indices
  - $\binom{M}{2} = \frac{M(M-1)}{2}$ second-order indices
  - $\binom{M}{k}$ k-th order indices
- The sum of all Sobol' indices (of any order) is 1

First-order indices

- $\sum_{i=1}^{M} S_i \leq 1$
- $\sum_{i=1}^{M} S_i = 1 \Leftrightarrow$ there are no interactions in the model (additive model)

Properties of Sobol' indices

- There are $2^M - 1$ different indices including:
  - M first-order indices
  - $\binom{M}{2} = \frac{M(M-1)}{2}$ second-order indices
  - $\binom{M}{k}$ k-th order indices
- The sum of all Sobol' indices (of any order) is 1

First-order indices

- $\sum_{i=1}^{M} S_i \leq 1$
- $\sum_{i=1}^{M} S_i = 1 \Leftrightarrow$ there are no interactions in the model (additive model)
- They quantify the additive effect of each input parameter separately, i.e., the reduction of variance to expect if X_i was set equal to a deterministic value

Properties of Sobol' indices

- There are $2^M - 1$ different indices including:
  - M first-order indices
  - $\binom{M}{2} = \frac{M(M-1)}{2}$ second-order indices
  - $\binom{M}{k}$ k-th order indices
- The sum of all Sobol' indices (of any order) is 1

First-order indices

- $\sum_{i=1}^{M} S_i \leq 1$
- $\sum_{i=1}^{M} S_i = 1 \Leftrightarrow$ there are no interactions in the model (additive model)
- They quantify the additive effect of each input parameter separately, i.e., the reduction of variance to expect if X_i was set equal to a deterministic value
- A large first-order index S_i means that X_i is important
- A small S_i does not mean that X_i is not important (interactions!)

Total Sobol' indices

Total Sobol' indices collect all effects of a variable (i.e., main effect and all interaction effects):

$$
\begin{aligned}
S_i^T &= S_i + \sum_{j \neq i} S_{ij} + \sum_{j,k \neq i} S_{ijk} + \dots \\
&= \frac{1}{D} \sum_{\boldsymbol{\alpha}:\alpha_i > 0} c_{\boldsymbol{\alpha}}^2
\end{aligned}
$$

Total Sobol' indices

Total Sobol' indices collect all effects of a variable (i.e., main effect and all interaction effects):

$$
\begin{aligned}
S_i^T &= S_i + \sum_{j \neq i} S_{ij} + \sum_{j,k \neq i} S_{ijk} + \dots \\
&= \frac{1}{D} \sum_{\boldsymbol{\alpha}:\alpha_i > 0} c_{\boldsymbol{\alpha}}^2
\end{aligned}
$$

Properties

- $S_i \leq S_i^T$
- $S_i = S_i^T \Leftrightarrow$ there are no interactions involving X_i
- $\sum_{i=1}^M S_i^T \geq 1$ (equality for additive models)
- S_i^T small $\Rightarrow X_i$ is non-influential and can be set to a deterministic value in subsequent analyses

Example

$(M = 2)$

Computational model $\quad Y = \mathcal{M}(X_1, X_2)$

Probabilistic model $\quad X_i \sim \mathcal{N}(\mu_i, \sigma_i)$

Isoprobabilistic transform $\quad X_i = \mu_i + \sigma_i\, \xi_i$

Chaos degree $\quad p = 3$, *i.e.* $P = 10$ terms

j	$\boldsymbol{\alpha}$	$\Psi_{\boldsymbol{\alpha}} \equiv \Psi_j$
0	$[0,0]$	$\Psi_0 = 1$
1	$[1,0]$	$\Psi_1 = \xi_1$
2	$[0,1]$	$\Psi_2 = \xi_2$
3	$[2,0]$	$\Psi_3 = (\xi_1^2 - 1)/\sqrt{2}$
4	$[1,1]$	$\Psi_4 = \xi_1\xi_2$
5	$[0,2]$	$\Psi_5 = (\xi_2^2 - 1)/\sqrt{2}$
6	$[3,0]$	$\Psi_6 = (\xi_1^3 - 3\xi_1)/\sqrt{6}$
7	$[2,1]$	$\Psi_7 = (\xi_1^2 - 1)\xi_2/\sqrt{2}$
8	$[1,2]$	$\Psi_8 = (\xi_2^2 - 1)\xi_1/\sqrt{2}$
9	$[0,3]$	$\Psi_9 = (\xi_2^3 - 3\xi_2)/\sqrt{6}$

Variance

$$D = \sum_{j=1}^{9} c_j^2$$

Sobol' indices

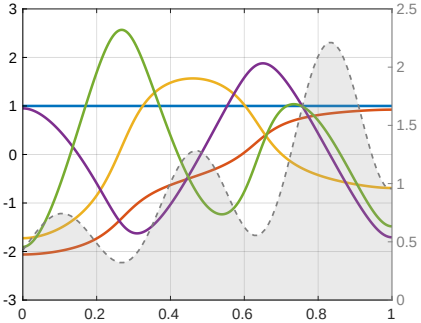
$$\begin{aligned} S_1 &= \left(c_{[1,0]}^2 + c_{[2,0]}^2 + c_{[3,0]}^2\right)/D \\ S_2 &= \left(c_{[0,1]}^2 + c_{[0,2]}^2 + c_{[0,3]}^2\right)/D \\ S_{12} &= \left(c_{[1,1]}^2 + c_{[2,1]}^2 + c_{[1,2]}^2\right)/D \end{aligned}$$

Activity: Compute the **first-order and the total Sobol' index** for X_2 for the sparse PCE from slide 10:

j	Multi-index $\boldsymbol{\alpha}$	Coefficient value
0	[0 0 0]	7.2
1	[0 1 0]	3.4
2	[0 0 1]	-0.9
3	[0 1 1]	0.1
4	[0 2 0]	1.7
5	[2 1 0]	-0.02
6	[0 3 0]	-0.2

Outlook: Using other chaos expansions for Sobol' indices

- In the derivation, only orthonormality of the basis was used
- It doesn't matter that the basis functions are polynomials!
- Other orthonormal bases allow the same way of computation (provided the 0-th basis function is constant)



$\rightarrow$ Poincaré chaos expansions (Part 3)

Practical exercise: Work on the corresponding Jupyter notebook to learn how to conduct PCE-based Sobol' analyses with UQ[PY]LAB.